Scheduling in Prolog

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Toronto Metropolitan University

CPS721: Artificial Intelligence I

Learning Objectives

Students should be able to

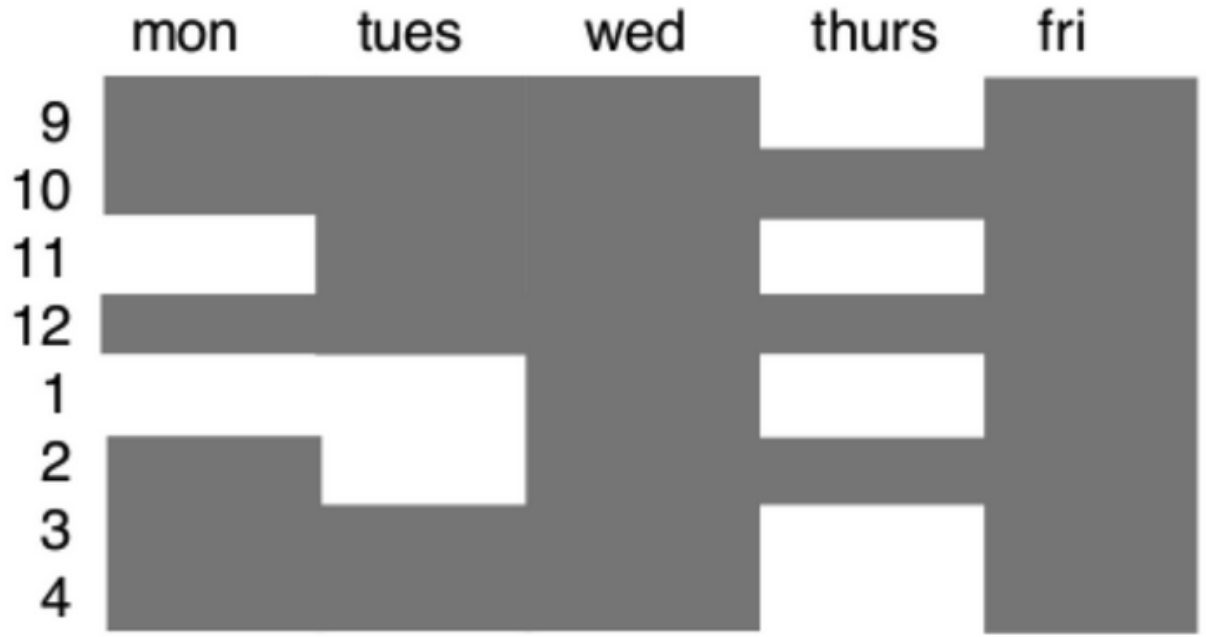
1. Encode a scheduling problem in Prolog

2. Use not to define certain constraints

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Meeting Scheduling

- Need to book 5 hours for meetings for the coming week
- Only one conference room is available, but already partially booked
- Conference room is booked in one hour from 9-5pm



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Constraints

- Can’t select already booked times
- Non-consecutive hours only
- No more than 2 hours on any day

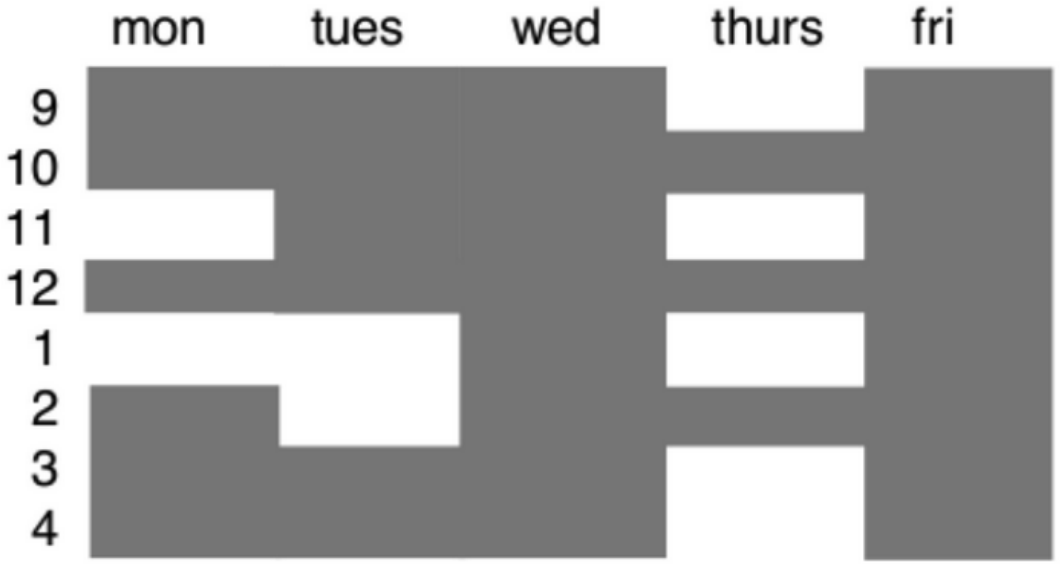
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Variables and Domains

- What are the variables and domains?

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Room Booked Constraint




```
taken([X, 12]).
taken([wed, X]).
taken([fri, X]).
```

```
taken([mon, 9]).
taken([mon, 10]).
taken([mon, 2]).
taken([mon, 3]).
taken([mon, 4]).
```

```
taken([tues, 9]).
taken([tues, 10]).
taken([tues, 11]).
taken([tues, 3]).
taken([tues, 4]).
```

```
taken([thurs, 10]).
taken([thurs, 2]).
```

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“No More than 2 Hours a Day”

- Often more convenient to implement the negation of a constraint
  - Then use not

```
threeOnOneDay(L) :-
```

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“Non-Consecutive Hours Only”

```
twoConsHours(L) :-
```

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Scheduling in Prolog

`taken` atomic statements

```
roomTime([D, H]) :- member(D, [mon, tues, wed, thurs, fri]),
      member(H, [9, 10, 11, 12, 1, 2, 3, 4]).

solve([T1, T2, T3, T4, T5]) :-
      roomTime(T1), roomTime(T2), roomTime(T3), roomTime(T4), roomTime(T5),
      not taken(T1), not taken(T2), not taken(T3), not taken(T4), not taken(T5),
      not twoConsHours([T1, T2, T3, T4, T5]),
      not threeOnOneDay([T1, T2, T3, T4, T5]).
```

Anything missing?

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Scheduling in Prolog

`taken` atomic statements

```
roomTime([D, H]) :- member(D, [mon, tues, wed, thurs, fri]),
      member(H, [9, 10, 11, 12, 1, 2, 3, 4]).

solve([T1, T2, T3, T4, T5]) :-
      roomTime(T1), roomTime(T2), roomTime(T3), roomTime(T4), roomTime(T5),
      not taken(T1), not taken(T2), not taken(T3), not taken(T4), not taken(T5),
      not twoCons([T1, T2, T3, T4, T5]),
      not threeOnOneDay([T1, T2, T3, T4, T5]),
      allDiff([T1, T2, T3, T4, T5]).
```

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Can We Improve This?

```
solve([T1, T2, T3, T4, T5]) :-
        roomTime(T1), roomTime(T2), roomTime(T3), roomTime(T4), roomTime(T5),
        not taken(T1), not taken(T2), not taken(T3), not taken(T4), not taken(T5),
        not twoConsecutiveHours([T1, T2, T3, T4, T5]),
        not threeOnOneDay([T1, T2, T3, T4, T5]),
        allDiff([T1, T2, T3, T4, T5]).
```

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